CSC 144-001 Discrete Mathematics for Computer Science I

MWF 9:00-9:50am, Ctr English 2nd Lng, Rm 102

Course Description

This course introduces the mathematics used in computer science. Logic and logical thinking, an area essential to computer science and many other subjects, is introduced. Formal definitions in logic are studied, but the ability to think and express logical ideas is key. Sets, relationships, and functions, the building blocks of many areas of mathematics are covered as well as practical areas of combinatorics, modular arithmetic, and discrete probability are covered. These are all skills that will serve a person well not only in computer science, but in many situations of life.

Catalog Description: The first of a two-course sequence introducing mathematical concepts for Computer Science. Topics include: sets, functions, and relations; propositional and predicate logic; modular arithmetic; and proofs.

Instructor and Contact Information

Instructor:

Eric Anson

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Tel: 520 621-2675

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(But note the **Course Communications** section below!)

TAs:

Miles Gendreau	milesgendreau@arizona.edu
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Class Email:

Questions about the class, assignments, or the material should be sent to the following email that goes to the instructor and all the Tas:

cs144s24@cs.arizona.edu

Office Hours

(Exact office hours subject to change, consult piazza for up-to-date information.)

Additional hours may be available by appointment.

Note: Office hours are periods of time we set aside to answer your questions. Please take advantage of them and come and talk with us.

Eric Anson

Monday 10:00-11:30

Wednesday 3:00-4:30

TAs Office Hours
TBA

Websites

Piazza: for questions https://piazza.com/arizona/spring2024/csc144001/home

D2L will be used for assignments, distributing grades and posting videos.

GradeScope will be used for turning in assignments and returning exams.

Course Format and Teaching Methods

There will be three lectures a week. Lectures will involve power point slides, input from students, and perhaps whiteboard work. Lectures will augment, enhance, and reiterate material from the textbook.

Course Objectives

- Learn to think logically and describe and understand how real word propositions can be written using propositional logic.
- Learn the different structures of proofs.
- Learn to write correct, well formatted proofs.
- Learn and understand the ideas behind sets, functions and relations.
- Learn some basic ideas of series, sequences.
- Learn some basic integer manipulation including modular arithmetic.

Expected Learning Outcomes

Students will be able to:

- Convert propositions from English to propositional logic and vice versa as well as evaluate the truth tables.
- Write clear, correct proofs of basic mathematical concepts.
- Explain with examples the basic terminology of functions, relations, and sets and perform

basic operations on these structures.

- Perform computations involving modular arithmetic.

Makeup Policy for Students Who Register Late

Students who register past the 1st day of classes will not be able to make up missed work. However, the policy of dropping the low quiz and assignment scores will allow a student who registers late to still get full points in the class.

Administrative Drops:

Every semester, students enroll in introductory CS classes but do not submit any work, resulting in a grade of 'E' at the end of the term. To prevent this, after the end of the second week, I will be administratively dropping all students who have not submitted any activities collected prior to the no-W drop date.

Course Communications

The best place to ask questions about dates, assignments or questions on the material should be through Piazza. More specific questions should be sent to the following email that goes to the instructor and all the Tas:

cs144s24@cs.arizona.edu

Websites

Piazza: for questions and discussion https://piazza.com/arizona/spring2024/csc144001/home

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Required Texts or Readings

Discrete Mathematics and Its Applications 8th edition, by Kenneth H. Rosen.

This is a required text and exercises will be assigned from it.

Assignments and Examinations: Schedule/Due Dates

Assignments

Written assignments will be given weekly on most weeks. These assignments must be completed, and the solutions scanned and turned in electronically on GradeScope. Late assignments will not be accepted, but the lowest two assignment grades will be dropped.

Pop Quizzes

In class quizzes will be given often. The dates of these quizzes will not be announced prior. Missed quizzes cannot be made up, but the grades on the lowest 1/4 of the quizzes will be dropped. In other words, if there are 8 quizzes, the lowest 2 quiz grades will be dropped. If there are 21 quizzes, the lowest 6 quiz grades will be dropped, etc.

Midterms

There will be three midterm exams. They will tentatively be held during the regular lecture times on Fri, Feb 16, Fri, Mar 22, and Fri Apr 26. Makeup exams will not be given, but the lowest score on a midterm exam will be replaced by the grade on the final exam if it is higher. Thus, if a midterm exam must be missed, the student will not receive a 0 on that exam.

Final

The final is scheduled for:

Thu, May 9 10:30am-12:30pm

Final Examination

The final is scheduled for:

Thu, May 9, 10:30am-12:30pm

Grading Scale and Policies

I use the common 90-80-70-60 grading scale.

The grades will be weighted as follows:

Quizzes	=	10%
Assignments	=	25%
Midterm Exams (15% each)	=	45%
Comprehensive Final Exam	=	20%
Total	=	100%

Homework will typically be graded within a week of when it is due. Exams will be graded at most 6 days after they are given. If exceptions must be made occasionally, staff will inform students about the delay and the reason given.

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at http://catalog.arizona.edu/policy/grades-and-grading-system#incomplete and http://catalog.arizona.edu/policy/grades-and-grading-system#Withdrawal respectively.

Dispute of Grade Policy

Any disputes about grades on assignments or exams must be submitted within one week of the grade being published.

Scheduled Topic and Activities

This is a list of the topics we will cover. The dates may change slightly. Every semester the pace moves a little differently based on student questions, etc.

Week 1	Jan 10	Introduction	
	Jan 12	Logic: Propositional Logic	(Chap 1.1)
Week 2	Jan 15	MLK Day (holiday)	
	Jan 17	Applications of Propositional Logic	(Chap 1.2)
	Jan 19	Propositional Equivalences	(Chap 1.3)
Week 3	Jan 22	Predicates and Quantifiers	(Chap 1.4)
	Jan 24	Predicates and Quantifiers	(Chap 1.4)

	Jan 26	Nested Quantifiers	(Chap 1.5)
Week 4	Jan 29	Rules of Inference	(Chap 1.6)
	Jan 31	Logic: Introduction to Proofs	(Chap 1.7)
	Feb 2	Logic: Introduction to Proofs	
Week 5	Feb 5	Logic: Introduction to Proofs	
	Feb 7	Logic: Proof Methods and Strategy	(Chap 1.8)
	Feb 9	Logic: Proof Methods and Strategy	
Week 6	Feb 12	Logic: Proof Methods and Strategy	
	Feb 14	Review/Buffer Day	
	Feb 16	Exam 1	
Week 7	Feb 19	Sets	(Chap 2.1)
	Feb 21	Set Operations	(Chap 2.2)
	Feb 23	Functions	(Chap 2.3)
Week 8	Feb 26	Functions	(Chap 2.3)
	Feb 28	Sequences and Summations	(Chap 2.4
	Mar 1	Cardinality of Sets	(Chap 2.5)
	Mar 2-10	SPRING BREAK	
Week 9	Mar 11	Cardinality of Sets	(Chap 2.5)
	Mar 13	Basic Structures: Matrices	(Chap 2.6)
	Mar 15	Divisibility and Modular Arithmetic	(Chap 4.1)
Week 10	Mar 18	Divisibility and Modular Arithmetic	(Chap 4.1)
	Mar 20	Review/Buffer Day	
	Mar 22	Exam 2	
Week 11	Mar 25	Primes and Greatest Common Divisors	(Chap 4.3)
	Mar 27	Integer Representations and Algorithms	(Chap 4.2)
	Mar 29	Primes and Greatest Common Divisors	(Chap 4.3)
Week 12	Apr 1	Applications of Congruences	(Chap 4.5)
	Apr 3	Cryptography	(Chap 4.6)
	Apr 5	Cryptography	(Chap 4.6)
Week 13	Apr 8	Relations: Relations and Their Properties	(Chap 9.1)
	Apr 10	n-ary Relations	(Chap 9.2)
	Apr 12	Representing Relations	(Chap 9.3)
Week 14	Apr 15	Closure of Relations	(Chap 9.4)
	Apr 17	Equivalence Relations	(Chap 9.5)
	Apr 19	Partial Ordering	(Chap 9.6)
Week 15	Apr 22	Partial Ordering	(Chap 9.6)
	Apr 24	Review/Buffer Day	
	Apr 26	Exam 3	
Week 16	Apr 29	TBA	
	May 1	TBA	

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and may be reported to the Dean of Students.

Some learning styles are best served by using personal electronics, such as laptops and iPads. These devices can be distracting to other learners. Therefore, students who prefer to use electronic devices for note-taking during lecture are welcome to do so, but if you plan to use the device for any other purpose (silently) such as web browsing, games, video, etc please sit in the last row so you don't distract fellow students.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): https://cirt.arizona.edu/case-emergency/overview

Also watch the video available at https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/common/learningeventdetail/crtfy000000000003560

University-wide Policies link *[Instructors may include these policies directly on their syllabus; providing the text above is sufficient]*

Links to the following UA policies are provided here, http://catalog.arizona.edu/syllabus-policies:

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy

Department-wide Syllabus Policies and Resources link *[Instructors may include these policies and resources directly on their syllabus; providing the text above is sufficient]*

Links to the following departmental syllabus policies and resources are provided here, https://www.cs.arizona.edu/cs-course-syllabus-policies :

- Department Code of Conduct
- Class Recordings
- Illnesses and Emergencies
- Obtaining Help
- Preferred Names and Pronouns
- Confidentiality of Student Records
- Additional Resources
- Land Acknowledgement Statement

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.